

Two Jug Problem

There is a large bucket B full of water and Two (02) jugs, J1 of volume 3 litre and J2 of volume 5 litre.

You are allowed to fill up any empty jug from the bucket, pour all water back to the bucket from a jug or pour from one jug to another.

The goal is to have jug J1 with exactly one (01) litre of water.

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State Definition: $\langle J1, J2 \rangle$

Start: $\langle 0, 0 \rangle$, Goal: $\langle 1, 0 \rangle$

Rules:

1. Fill (J1): $\langle J1, J2 \rangle$ to $\langle 3, J2 \rangle$
2. Fill (J2): $\langle J1, J2 \rangle$ to $\langle J1, 5 \rangle$
3. Empty (J1), Empty (J2): Similarly defined
4. Pour (J1, J2): $\langle J1, J2 \rangle$ to $\langle X, Y \rangle$, where
 - a. $X = 0$ and $Y = J1 + J2$ if $J1 + J2 \leq 5$,
 - b. $Y = 5$ and $X = (J1 + J2) - 5$, if $J1 + J2 > 5$
5. Pour (J2, J1): Similarly defined

